

Resonant states reveal strong light-matter coupling in nanophotonic cavities

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Photonic resonances are a powerful tool for controlling light-matter interactions. However, unlocking many of the most scientifically intriguing and technologically promising phenomena requires entering the strong coupling regime, where light and matter fully mix, unlocking emergent properties of the coupled states. Nowadays, distinguishing between weak and strong coupling primarily relies on studying the optical response of the hybrid system at real frequencies, which only provides indirect estimates of the underlying resonant dynamics. In contrast, the actual resonances live at complex frequencies. Resolving this contradiction, we show that photonic resonant states provide the framework to unambiguously quantify the strength of light-matter interaction, enabling a rigorous distinction between weak and strong coupling regimes. Assuming a single dominant resonant state of the bare photonic resonator, we derive an effective Hamiltonian that captures the interaction between the photonic resonator and an arbitrary number of material resonances. Our analysis reveals that, unlike most coupled-oscillator models commonly employed in the literature, hybridization not only introduces off-diagonal coupling but also shifts the bare eigenfrequency of the photonic mode. We demonstrate the accuracy of this approach by studying planar and spherical silver resonators filled with a molecular material whose properties were extracted from quantum-chemical simulations. Our work paves the way towards a unified description of light-matter coupling in open photonic environments.

I. INTRODUCTION

Hybrid states of light and matter form when photonic and material resonances are coupled [see Figure 1] [1, 2]. When the coupling is strong, the photon and material resonances each contribute a significant part to the resulting hybrid modes, instead of just perturbing one another. These so-called polaritons possess emergent properties distinct from their constituents [3]. As such, they enable the development of advanced technologies, comprising efficient polariton-lasers [4, 5], all-optical compute [6–9], optical neural networks [10], and high-resolution (label-free) sensors [11, 12]. Overall, the range of possible applications leveraging polaritons is vast, spanning from solar cells and LEDs [13] to platforms for topological physics [9]. Recent advances in polaritonic chemistry have even shown that strong coupling can be leveraged to modify the reaction energy landscape to favor selected reaction pathways [14–16].

Typically, these applications benefit from an increased coupling strength between photonic and material resonances. To assess the strength of light-matter coupling in optical systems, a standard approach is to compare the spectra of the uncoupled components — the photonic resonator and the material — with that of the coupled system. In the strong coupling regime, spectrally aligned bare-system resonances typically split into two spectrally separated peaks in the coupled spectrum. The spectral separation between these split peaks is widely

used as a measure of the coupling strength. However, accurately assessing the coupling strength is subtle, as the inherent non-Hermitian nature of open photonic systems, that can be probed from outside, blurs the view of the resonances [17–20]. For instance, the authors of Reference [17] showed how the splitting extracted from peaks/dips in the spectra of different observable quantities of the same system can differ widely. Moreover, many realistic materials feature multiple closely spaced material resonances [19, 21, 22] [see Figure 1(c)], adding further complications to the extraction of coupling parameters from observable spectra. The fundamental problem is that when judging the coupling strength based on an optical quantity defined at a real frequency, we only indirectly probe the features of the resonances. The resonances in most open systems live at complex frequencies, and ultimately, the question of whether strong coupling emerges or not requires a consideration of the resonance interaction at these complex frequencies.

In this work, we distinguish weak and strong light-matter coupling at complex frequencies. Our analysis is based on the resonant states (RSs) — also known as quasi normal modes, decaying states, natural modes, etc. — constituting a natural framework to describe photonic resonances. Unlike probing the optical response under some illumination at real frequencies, the study of the RSs allows an explicit determination of the resonance frequencies and the damping rates of the hybrid structure, without the need for any fitting to optical spectra. In contrast to previous studies [18, 23–26], we derive for the first time an effective Hamiltonian that explicitly shows the degree of mixing between the material resonances and

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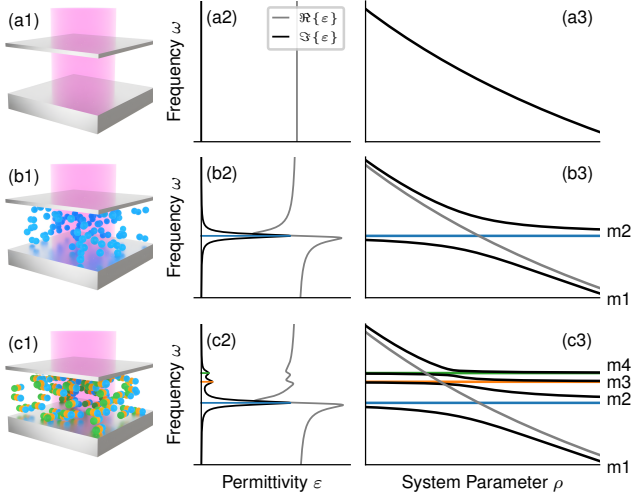


FIG. 1. Mode hybridization: (a) When a system parameter ρ of a photonic resonator, which can be considered as an independent oscillator, is changed, the system dynamics change accordingly, expressed here in terms of a changing (complex) eigenfrequency as a function of ρ . Examples for possible choices of ρ are the spacing between the plates for a Fabry-Perot cavity or the radius of a core or a shell for a spherical resonator. (b) When a material characterized by a single Lorentz-oscillator is introduced to the photonic resonator, the photonic mode and material resonance couple to each other, leading to hybridization. The new hybrid modes distribute their energy across the coupled system, leading to shifts in the eigenfrequencies from the bare system without coupling. (c) The simple relation between coupling coefficients and frequency splitting gets spoiled when multiple material resonances are introduced. Column (1) illustrates the considered system, column (2) expresses the material properties, and column (3) shows the (un-)coupled eigenfrequencies.

a single RS of the bare photonic resonator directly from macroscopic Maxwell's equations, yielding a closed-form expression for the coupling rate. Crucially, we discover that, unlike conventional coupled-oscillator models, light-matter coupling also introduces an additional spectral shift of the bare photonic eigenfrequency.

We will start our discussion with a brief review of the theoretical background on Maxwell's equations, resonant states, pole-based permittivity models, and the coupled oscillator model. We will briefly discuss various criteria considered to classify whether a system exhibits strong coupling. A commonality between the criteria is the necessity to accurately evaluate the coupling rate. We go on to provide a RS formulation of the coupling between a single RS and multiple material resonances.

We will then proceed to investigate example systems of increasing complexity. First, we will treat a planar cavity filled with a medium characterized by a single Lorentz-resonance. It serves to discuss how signatures of strong coupling are obscured by radiative and dissipative damping. Using the same example, we will show how RSs allow us to discriminate between weak and strong coupling,

even when optical observables yield contradictory conclusions. In our second example, we analyze a core-shell nanosphere, showing the validity of our approach for single nanoparticles. For the first time, our method enables direct access to the individual coupling rates without the need for phenomenological coupled-oscillator models. We go on to demonstrate how these rates are particularly useful when investigating systems that host multiple adjacent material resonances. In such systems, the overall splitting of the hybrid modes results from complex contributions of the different material resonances. With our method, these contributions can be disentangled from each other.

Our findings establish RSs as a clear and universal framework for light-matter coupling in photonic resonators, avoiding ambiguous spectral criteria and paving the way to systematically design and engineer strong coupling in complex photonic systems.

II. RESULTS

A. Theoretical background

The propagation of electromagnetic waves in the frequency domain is governed by macroscopic Maxwell's curl equations [27]:

$$\begin{aligned}\nabla \times \mathbf{H}(\mathbf{r}, \omega) &= -i\omega \mathbf{D}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega), \\ \nabla \times \mathbf{E}(\mathbf{r}, \omega) &= i\omega \mathbf{B}(\mathbf{r}, \omega),\end{aligned}\quad (1)$$

where $\mathbf{E}(\mathbf{r}, \omega)$ and $\mathbf{B}(\mathbf{r}, \omega)$ are the electric and magnetic fields, respectively, while $\mathbf{D}(\mathbf{r}, \omega)$ and $\mathbf{H}(\mathbf{r}, \omega)$ are auxiliary fields. For readability, we will assume isotropic non-magnetic materials, *i.e.*,

$$\begin{aligned}\mathbf{B}(\mathbf{r}, \omega) &= \mu_0 \mathbf{H}(\mathbf{r}, \omega), \\ \mathbf{D}(\mathbf{r}, \omega) &= \varepsilon_0 \mathbf{E}(\mathbf{r}, \omega) + \mathbf{P}(\mathbf{r}, \omega),\end{aligned}\quad (2)$$

where the polarization $\mathbf{P}(\mathbf{r}, \omega) = \varepsilon_0 \chi(\mathbf{r}, \omega) \mathbf{E}(\mathbf{r}, \omega)$ in the material is linearly related to the electric field by the susceptibility $\chi(\mathbf{r}, \omega)$. ε_0 and μ_0 are the vacuum permittivity and permeability, respectively.

Describing resonant light-matter interaction naturally calls for the introduction of RSs as the mathematical manifestation of photonic resonances. These RSs are defined as solutions of the source-free Maxwell equations with radiation boundary conditions. Strictly speaking, RSs also comprise optical resonances that are decoupled from the radiation continuum, known as bound states in the continuum [28]. However, these RSs cannot be accessed by far-field measurements. For all other RS, the openness of the system necessarily leads to damping over time. In contrast to eigenmodes of closed systems or bound states in the continuum, the eigenfrequencies of RSs are thus complex-valued, with the imaginary part determining the exponential damping rate in time. As a consequence, the eigenfields diverge away from the

resonator, which results in several challenges, including their normalization, orthogonality, and completeness [29–31]. If an observable existed that was coupled solely to a single RS with eigenfrequency $\tilde{\omega}_c = \omega_c - i\gamma_c$, the observable would exhibit a Lorentzian lineshape centered around the real part of that eigenfrequency (we call ω_c its resonance frequency) and broadened according to the imaginary part of its eigenfrequency (we call γ_c its damping rate). However, RSs can interfere with each other and with a non-resonant background, forming intricate Fano lineshapes [20, 32–35].

Now that we have laid the foundation for the treatment of electromagnetic resonances, let us switch our attention to the description of resonant material excitations. While a rigorous *ab-initio* description requires solving the interaction of light and matter quantum-mechanically, semi-classical descriptions boil down to the movement of bound charges q (with effective mass m_0) that are displaced from their equilibrium positions in the effective potential landscape of the inner electrons and the atomic nuclei by $\mathbf{x}(t)$ due to the presence of a driving field. Solving the corresponding differential equations yields the induced polarization $\mathbf{P}(t) = Nq\mathbf{x}(t)$ with number density N . In the frequency domain, this results in the well-established Lorentz model [36]:

$$\begin{aligned}\varepsilon_r(\omega) &= 1 + \chi(\omega) \\ &= 1 + \frac{Nq^2}{m_0\varepsilon_0} \sum_j \frac{f_j}{\omega_{0,j}^2 - \omega^2 - i\Gamma_j\omega}.\end{aligned}\quad (3)$$

Here, f_j is the strength of the j^{th} oscillator with resonance frequency $\omega_{0,j}$ and damping Γ_j . Additional Drude-terms can be added to account for unbound charges [37].

To unravel the contribution of each material resonance, we write Equation (3) as a sum of resonant contributions [38]

$$\varepsilon_r(\omega) = \varepsilon_\infty + \sum_p \frac{i\sigma_p}{\omega - \Omega_p}, \quad (4)$$

where σ_p corresponds to the excitation strength of the p^{th} material resonance, Ω_p is the complex frequency of the material-pole, and ε_∞ encapsulates the background permittivity resulting from resonances far outside the frequency range of interest. For convenience, we provide the expressions for σ_p and Ω_p linking Equation (4) to Equation (3) in the Supplementary Information (SI).

B. Criteria for strong coupling in open systems

To gain first insights into the physics of light-matter coupling, we will first consider a standard phenomenological model of two coupled harmonic oscillators. As open systems are inherently lossy, we will include a damping in the form of complex-valued initial (bare) eigenfrequencies of the optical cavity $\tilde{\omega}_c = \omega_c - i\gamma_c$ and the material resonance: $\Omega_p = \omega_p - i\gamma_p$. The eigenfrequency of the

cavity mode $\tilde{\omega}_c(\rho)$ is assumed to be a function of a system parameter ρ — such as the cavity thickness in a Fabry-Perot (FP) cavity. The coupled equations of motion describe the joint dynamics due to the light-matter ($\hat{\kappa}$) and matter-light (κ) coupling terms (see [18] when $\kappa = \hat{\kappa}$):

$$\underbrace{\begin{pmatrix} \Phi_c(\rho) \\ \Phi_p(\rho) \end{pmatrix}}_{\Phi(\rho)} = \underbrace{\begin{bmatrix} \tilde{\omega}_c(\rho) & \kappa(\rho) \\ \hat{\kappa}(\rho) & \tilde{\Omega}_p \end{bmatrix}}_{\mathcal{H}} \begin{pmatrix} \Phi_c(\rho) \\ \Phi_p(\rho) \end{pmatrix}. \quad (5)$$

Here, $\Phi_c(\rho)$ and $\Phi_p(\rho)$ are coefficients quantifying the contributions of the cavity mode and material resonance to the hybrid system [16]. For readability, the ρ dependence will not be stated explicitly throughout the rest of the derivation. By diagonalizing $\mathcal{H}$, we obtain uncoupled equations of motion for superpositions of the initial modes (considering the material resonance as a further mode of the system). These quasi-particles are commonly referred to as polaritons. They evolve with distinct eigenfrequencies

$$\tilde{\omega}_{1,2} = \frac{\tilde{\Omega}_p + \tilde{\omega}_c}{2} \pm \frac{1}{2} \sqrt{4\hat{\kappa}\kappa + (\tilde{\Omega}_p - \tilde{\omega}_c)^2}.$$

On resonance ($\omega_c = \omega_p = \omega_0$), we get

$$\begin{aligned}\tilde{\omega}_{1,2} &= \omega_0 - \frac{i(\gamma_p + \gamma_c)}{2} \pm \frac{\Omega_R}{2}, \\ \text{with } \Omega_R &:= \sqrt{4\hat{\kappa}\kappa - (\gamma_p - \gamma_c)^2}.\end{aligned}\quad (6)$$

For real-valued $\hat{\kappa}\kappa$, *i.e.*, retardation-free coupling, and

$$\hat{\kappa}\kappa > \frac{1}{4}(\gamma_p - \gamma_c)^2, \quad (7)$$

the splitting Ω_R is real-valued. Here, we consider Equation (7) to mark the transition from **weak** to **(hidden) strong coupling**. **Damping-induced linewidth broadening obscures the splitting of resonance frequencies to a point that it may not be observed in typical experimental measurements, leading us to introduce the term "hidden".** Note how at the same time, the existence of damping is a prerequisite for the existence of the weak-coupling regime. To compensate for the effects of damping, a collection of alternative criteria for strong coupling has been used throughout the literature [16, 27, 39–48], which will be further discussed in a dedicated section. Ω_R is commonly denoted as Rabi frequency, analogous to Rabi oscillations in optically-excited atoms. It can be understood as the frequency at which energy is exchanged between the coupled modes. Figure 1(b) illustrates the resulting two-branched dispersion, featuring an avoided crossing with a separation corresponding to Ω_R . Notice how the spectral separation increases with detuning sufficiently far from resonance, when $|\tilde{\Omega}_p - \tilde{\omega}_c(\rho)|$ dominates over κ in the discriminant. In experimental observations, where only the dynamics of the coupled system are accessible, it is thus common to assume that the resonance

frequency and the frequency of minimum separation between the upper and lower polariton branches (UP/LP) coincide. We remark that in cases where the coupling parameter $\kappa(\rho)$ and/or cavity loss $\gamma_c(\rho)$ change significantly with ρ , the minimum separation between UP and LP is shifted away from the configuration ρ for which the real parts of the bare eigenfrequencies coincide [49].

C. Observable versus “hidden” strong coupling

The above treatment already considered loss channels originating from the photonic resonator and dissipation in the material. However, their implications on observable physics remain to be clarified. Effects relying on the coherent exchange of energy between modes are suppressed when the decay dominates. Applications in quantum information are particularly vulnerable to the resulting infidelity [50]. In classical experiments, the spectral position of the resonant modes is most often determined from peaks or dips in spectrally-resolved observables such as reflection and transmission, or derived quantities such as absorption and, in the case of a few microscopic (plasmonic) resonators, scattering-, extinction-, and absorption-cross-sections. In these measurements, the predominant fingerprint of considerable loss $\gamma_{p,c}$ is a broadening of the observed line shape. As a consequence, the avoided crossing of coupled RSs that fulfill the strong-coupling criterion [Equation (7)] can be experimentally indiscernible. This motivates the introduction of an additional regime, when the coupling rate dominates over all present loss channels. Henceforth, we consider a system with one material resonance to exhibit **observable strong coupling** if, in addition to Equation (7), the hybrid modes are sufficiently separated to be discernible despite their broadening.

The discernibility of adjacent broadened peaks is a lively field of ongoing scientific activity [51–53]. As a result, the criterion for observable strong coupling is less clearly defined than the transition from weak to strong coupling. For practicality, we deem superimposed Lorentzian lineshapes discernible if their centers are separated further than the sum of their intensity half widths at half maximum (γ), which translates to the criterion (considering the factor $\frac{1}{2}$ in the definitions of γ , see [27, 39–42]):

$$\frac{\Omega_R}{2} > \frac{\gamma_p + \gamma_c}{2}. \quad (8)$$

Using the definition in Equation (6), it is possible to rewrite this criterion in terms of the coupling coefficients $\hat{\kappa}\kappa$ [27, 41]:

$$\hat{\kappa}\kappa > \frac{\gamma_p^2 + \gamma_c^2}{2}. \quad (9)$$

Unfortunately, optical measurements (such as transmission or reflection measurements) do not provide di-

rect access to the damping rates and the coupling coefficients that enter Equation (7) and Equation (9). This is especially true for rich spectra with more than two resonances in the same spectral region, making interpretation challenging regardless of whether the coupling is weak, strong, or observably strong. As discussed next, there is even an ambiguity in the extraction of Ω_R from observable spectra. Nonetheless, it is the established practice to find these parameters from fitting effective coupled-oscillator models to experimental or simulated spectra [27, 39, 40].

D. From weak to observable strong coupling

To motivate the need to use RSs in the investigation of light-matter coupling, let us start by analyzing a simplified system featuring a single material pole coupled to a Fabry-Perot mode of a planar cavity. The details of the system are summarized in Table I of the SI. For illustrative purposes, we will operate the system in the three coupling regimes and traverse from *weak coupling* to *strong coupling*, and finally to *observable strong coupling*. To achieve such a transition between the different coupling regimes, we artificially reduce the oscillator strength f_j or equivalently the residue of the corresponding material pole $\sim \sigma_p$ — and, with it, increase the coupling between light and matter — by a factor η of 0.025, 0.05, 0.1, and 1, respectively.

In an attempt to study the coupling regime, one would typically collect spectra of some observable quantity. We illustrate this approach by considering scattering matrix based thin film calculations of the cavity under normal incidence from the side with the thinner silver layer. In Figure 2(a1-a4), we show the transmission T in black. Two peaks separated by a valley are present for all η . With increasing coupling strength, the frequency separation between the peaks increases. We want to emphasize that the presence of such a splitting does not prove the existence of strong coupling, as previously noted, *e.g.*, in Reference [16]. In particular, the splitting observed in Figure 2(a1) results from Fano-type modal interference. It is readily understood from a basic model [20]: The narrow molecular absorption line carves a dip into the broad transmission peak of the Fabry-Perot mode, leaving the impression of a splitting.

To shed light on the origin of this behavior, we foreshadow the RS-based analysis introduced later. The orange and blue lines in Figure 2(a1-a4) show the two dominant RS-contributions to the transmission. In the weak coupling regime (the first column of Figure 2 is later identified as weak coupling), these two contributions are nearly indistinguishable from the lineshape of the bare material resonance (orange) and of the bare FP mode (blue). In isolation, the material resonance is significantly narrower than the FP mode. Interference between these two RSs leads to the seemingly split transmission spectrum.

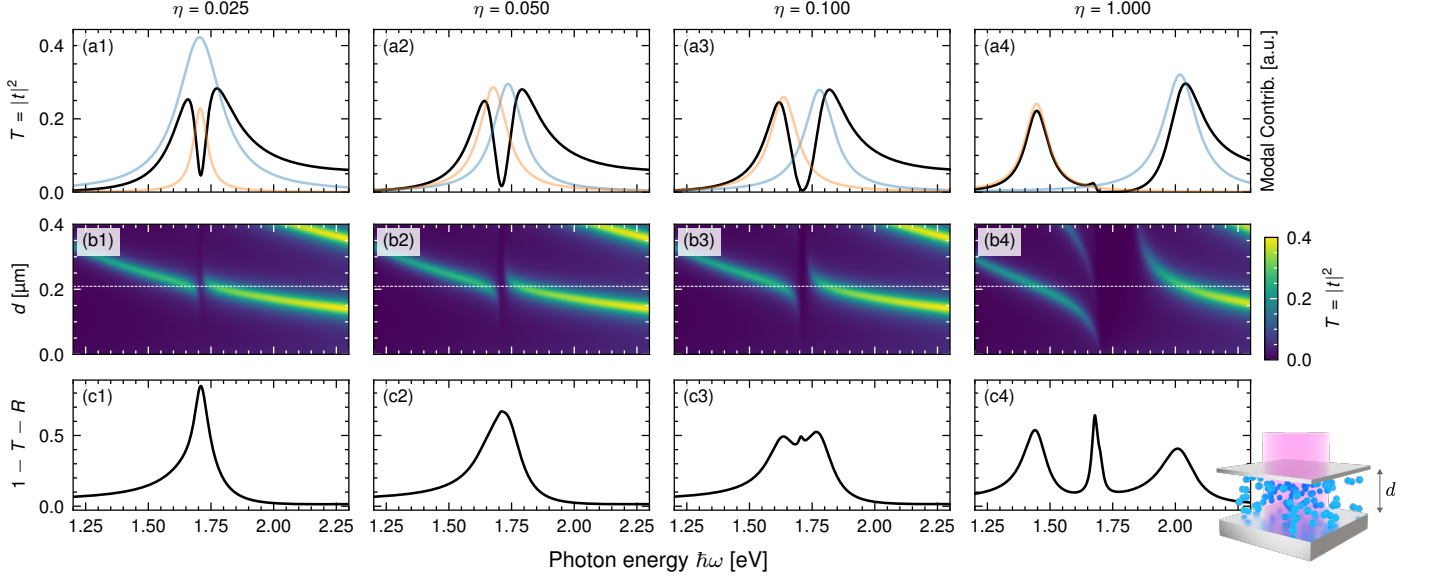


FIG. 2. Optical response of a planar cavity filled with a resonant medium in terms of observables commonly used as hallmarks of strong coupling. From left to right (1-4), the coupling strength is changed by scaling the oscillator strength of an artificial single-pole Lorentz medium by a factor η (material and geometry parameters are given in Table I). (a) The solid black line shows the spectrally resolved transmission through the cavity upon illumination from below. For illustrative purposes, the contributions of the two dominant RSs are indicated proportional to $|\omega - \tilde{\omega}_m|^{-2}$ (orange and blue lines), which corresponds to the Lorentzian line shapes the RSs would have in isolation. (b) The cavity mode is tuned by changing the cavity thickness d . Note how (b1) resembles an avoided crossing, and (a1) correspondingly shows two clearly separated peaks, despite the system being in the weak coupling regime. The perceived splitting results from the interference between two modes that are at the same real frequency but have different linewidths. This phenomenon is known in the framework of Fano resonances [20]. The broad cavity mode (increasing transmission) can be understood as the background, on which a narrow absorption dip arises due to the material resonance. (c) Absorption upon the same illumination as in (a).

With increasing coupling strength, the observed peaks increasingly correspond to individual RSs, which becomes clear considering the good spectral alignment between the contributions from the individual RSs with the full optical response in Figure 2(a4). The only indication of the broken correspondence between peaks and individual RSs at low coupling strengths is the non-Lorentzian lineshapes of the transmission [see Figure 2(a1-a3)].

As a remedy, the authors of reference [16] suggest measuring the spectrum as a function of some changing system parameter and observing the avoided crossing of the polariton branches (as schematically illustrated in Figure 1 for a varying system parameter ρ). In the current example of a Fabry-Perot cavity, we vary the cavity thickness d . The d -resolved transmission is shown in Figures 2(b1-b4). Figures 2(b3-b4) clearly display an avoided crossing, which is consistent with our later analysis, classifying these cases as *observable strong coupling*. However, Figures 2(b1) and (b2) also resemble an avoided crossing, which turns out to be a *weak coupling* and a *hidden strong coupling*, respectively. Different observable quantities, such as the Purcell factor, photoluminescence, or derived quantities, such as the absorption, depicted in Figures 2(c1-c4), will exhibit peaks at different energies [17, 18, 48]. Such an ambiguity necessitates a unified way of assessing the coupling between light and matter.

In the following, we will take such an approach using the RSs, which are intrinsic to the resonator — they do not depend on the external excitation.

We performed the same analysis as in Figure 2(b), but tracked the evolution of the RSs instead of the observable spectra. The scattering matrix based solver for layered media allows us to evaluate the scattering matrix at complex frequencies. We use a selective domain subdivision scheme [54] with the AAA algorithm [55, 56] and iterative sample refinement [57] to locate the poles of the determinant of scattering matrix, corresponding to the RSs. Figures 3(a1-a4) show how the real part of the eigenfrequencies (*i.e.* the resonance frequency) evolves with d . RSs that originate from the same bare cavity mode (*i.e.* FP mode) are shown in matching colors (blue, orange, green, and red). Only for the fundamental FP mode both hybrid modes (blue lines) fall within the plotting-window. Of the higher-order modes (orange, green and red) only one branch is present close to material resonance, while the other branch is far detuned. For now, we will focus our discussion on the RSs related to the fundamental mode. In Figure 3(a1) the highly dispersive mode consists predominantly of the fundamental FP mode, and the less dispersive mode is dominated by the material resonance. It is immediately clear that these two RSs cross in Figure 3(a1), while they avoid the cross-

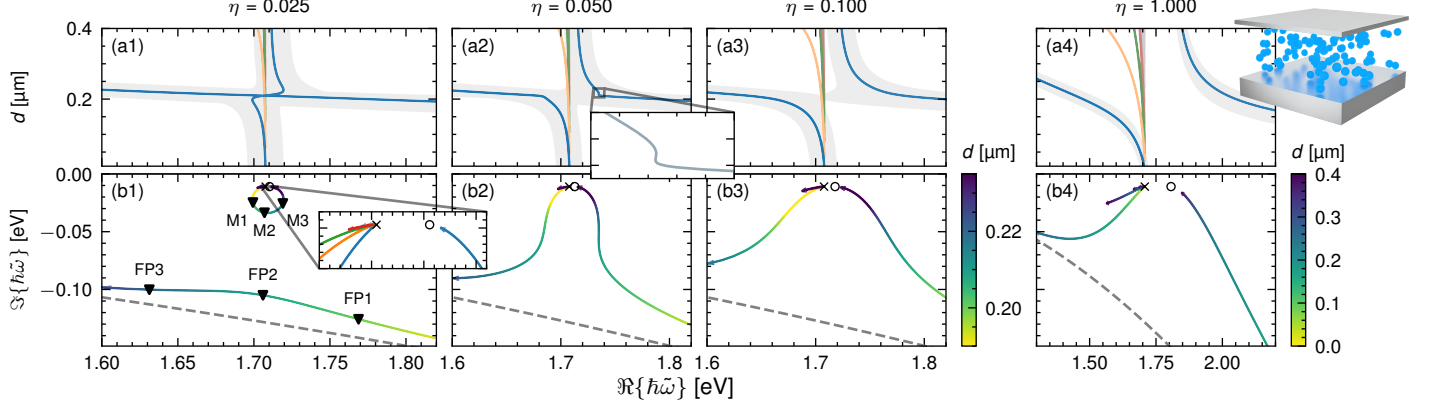


FIG. 3. Parametrized pole trajectories: The thickness of a planar cavity containing a single-pole Lorentz medium is changed, while the RS-eigenfrequencies are traced. The specific material parameters and geometry are given in Table I. As in Figure 2, the panels 1-4 correspond to increasing oscillator strength (scaled by η) of the material resonance, and thus increasing coupling strength. (a) The dispersion of the resonance frequencies (i.e., the real part of the eigenfrequencies) with changes in d . This provides a more precise view of the hybrid modes lifting the fog of damping-induced linewidth broadening (compare to Figures 2(b1-b4)). Lines of different colors correspond to cavity modes of different orders (blue being the fundamental mode). Each panel contains two branches, corresponding to distinct RSs resulting from the hybridization. In (a1) these branches cross, while in (a2-a4) they avoid the crossing. The correspondence to Figures 2(b1-b4) is further emphasized by the gray shaded region, covering $\Re\{\hbar\tilde{\omega}\} \pm \Im\{\hbar\tilde{\omega}\}$ of the fundamental mode, which can be understood as the linewidth of the corresponding Lorentzian. (b) The trajectories of the eigenfrequencies in the complex frequency plane as a function of the cavity thickness d (indicated by the color of the lines). The dispersion of the RSs associated with the lowest-order cavity mode is most prominent, as the eigenfrequency varies strongly within the chosen thickness interval. At the same time, the RSs associated with higher-order cavity modes barely move, resulting in short line segments. These segments only have an appreciable length in subfigure (a4) due to the larger interval in which the cavity thickness d is varied. The inset in (b1) shows additional details making the higher-order RSs more clearly visible (same color code as in the top row). This perspective of the complex plane clearly visualizes the attraction and repulsion of the hybrid modes in the weak and strong coupling regime, respectively. The triangle markers in (b1) are used in the main text to refer to sections of the trajectories when explaining these phenomena. FP and M denote hybrid modes that are dominated by Fabry-Pérot and material contributions, respectively. The dashed line indicates the dispersion of the bare cavity mode. A black cross (x) indicates the position of the single resonance of the Lorentz permittivity. A circle (o) marks its complex zero. RSs associated with higher-order cavity modes appear close to the material resonance, where an infinite number of modes are supported due to the diverging permittivity forming an accumulation point. The peculiar feature in the upper polariton branch in subfigure (b2; magnified in inset) results from the vicinity to the transition from weak to strong coupling, as it causes the eigenfrequency to locally vary almost exclusively in its imaginary part with changing d .

ing for (a2-a4), which corresponds to the transition from *weak* to *strong coupling* according to Equation (7). For $\eta = 0.025$ [Figure 3(a1)], the RSs cross because their difference in damping outweighs their strongly suppressed mutual coupling.

Figures 3(b1-b4) introduce an additional perspective by tracing the evolution of the RSs through the complex frequency plane with changing d . As such, Figures 3(a1-a4) can be understood as the projection of Figures 3(b1-b4) onto the real frequency axis. The pole trajectories in Figure 3(b1) reveal how the weakly-coupled FP mode modifies the material resonance. Together, they form hybrid states that are still clearly associated with either the FP mode or material resonance, respectively. This becomes apparent considering the asymptotic behavior when the cavity is far detuned to both extremes: The lower RS approaches the resonance of the photonic cavity (dashed line) while the other approaches the material pole (black cross) or zero (black circle) for small and large

cavity thickness, respectively.. Henceforth, we refer to them as quasi-FP (qFP) and quasi-material (qM) modes, respectively. For small d , the qFP mode is blue-detuned from the material resonance [FP1; triangle marker in Figure 3(b1)], while the coupling red-shifts the qM mode (akin to a Lamb shift; M1). When the real part of both modes matches (FP2; M2), predominantly the damping rate is modified (analogous to Purcell enhancement). After the qFP mode has passed by (FP3) upon increasing d even further, the qM mode gets blue-shifted (Lamb shift; M3).

After the transition to *strong coupling* [Figure 3 panel 2], the character of the hybrid modes is no longer clearly defined. Indeed, we see an entirely different behavior. The resonance frequency of one hybrid mode no longer crosses the other when sweeping the system parameter. Instead, we clearly observe that the hybrid modes smoothly exchange places in the complex frequency plane throughout the change of d .

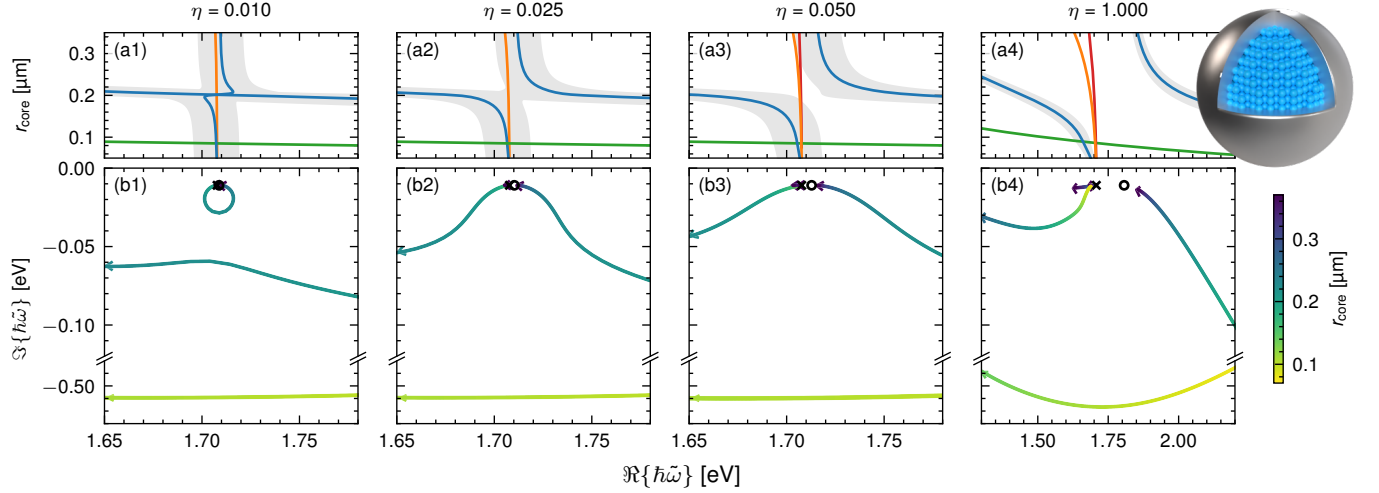


FIG. 4. Parametrized pole trajectories for a 3D finite system. In analogy to Figure 3, the evolution of the transverse magnetic dipolar resonances of a nanoparticle (NP) are traced. The NP consists of a SURMOF core with variable radius r_{core} and a fixed 20 nm silver shell. The plasmon resonance predominantly localized on the outer surface of the metallic shell [(bright) green line at the bottom of each panel] has vastly higher damping than the material resonance. The high damping results from radiation loss as the plasmon couples strongly to free space. At the same time it has low overlap with the SURMOF core. As a consequence, their interaction is negligible. In contrast, the other mode is predominantly localized at the inner surface of the metallic shell, characterized by a lower radiation loss and a better overlap with the SURMOF core. It is that mode that can be strongly coupled to the material excitation. As the light-matter coupling is varied across the columns, the plasmon mode predominantly localized at the inner surface of the shell [blue/teal lines in (a)/(b) respectively] traverses from the *weak* to the *observable strong coupling* regime.

While the hybrid mode on the right (high energies) starts as a mostly FP-type mode for a small cavity thickness d , it ends up approaching the material zero (indicated by the black circle) for large d . At the same time, a RS progressively emerges from the material pole (black cross) for a small thickness d and approaches the trajectory of the bare FP mode for large d (dashed line). Corresponding to Equation (7), the emergence of this distinct behavior is abrupt. Instead of just mutually perturbing one another, the hybrid modes exchange places, which constitutes an unambiguous signature of *strong coupling*. Exactly at the transition, the two RSs coalesce, marking an exceptional point, where the square root in Equation (6) vanishes. Similarly, the authors of References [24, 58] observed the transition from weak to strong coupling of two geometrically coupled optical RSs (far from any material resonance) through an exceptional point.

At $\eta = 0.05$ [Figure 3(2)], the system is still quite close to the transition between *weak* and *strong coupling*, which leads to a section of the trajectory, for which the resonance frequency — *i.e.* the real part of the complex eigenfrequency — is almost unchanged [seen in the near-vertical movement of the line in Figure 3(b2)]. In the projection onto the real axis [Figure 3(b2)], this leads to a peculiar feature in the modal dispersion, as highlighted in the inset.

From Figure 3(a), it becomes clear, however, that the loss-induced linewidth broadening obscures the emerging

avoided crossing. The broadening is illustrated by the gray-shaded regions that span $\omega \pm \gamma$ for each RS, and that merge at the point of the avoided crossing. Equation (8) is fulfilled as soon as there is no overlap between the gray regions associated with the different polariton branches. For $\eta = 0.1$ (panels 3 in Figures 2 and 3), this condition is on the verge of being met. Note how it coincides with the appearance of discernible peaks in the absorption spectrum [Figure 2(c3)]. In addition, a third peak emerges and becomes more pronounced for $\eta = 1$ (panels 4). The origin of this third peak is the subject of ongoing debate [59–62]. From the spectral analysis in Figure 2(b4) and the modal dispersion of RSs in Figures 3(a4) and (b4), it becomes apparent that the central peak originates from higher-order cavity modes. In particular, the higher-order modes of the bare cavity are highly detuned from the material resonance, resulting in hybrid modes that closely resemble the bare modes. As a consequence, their material branches sit very close to the original material resonance, jointly creating the central peak. In the complex plane, this collection of higher-order optical modes corresponds to an **accumulation point** of RSs at the material resonance, where the permittivity of the Lorentz model diverges. As a result, modes of arbitrarily high order (*i.e.*, short wavelength in the medium) fit into the cavity. Such accumulation points are visible in Figures 3 to 5.

Before investigating systems with multiple material resonances, let us briefly shift attention to a system of

finite size to highlight that the observed behavior is fundamental and not linked to a particular instance of a specific resonator. In this second example, we choose a spherical resonator — specifically a core-shell nanoparticle (NP). The core radius is chosen as the system parameter ρ to tune the properties of the bare system. Material and geometry parameters are again given in Table I in the SI. To find the RSs of the NP we search for the eigenvalues of the characteristic matrix, by a combination of the contour integral method and an iterative refinement. The core of the NP is made of the same artificial material as in the previous example with optical properties that are characterized by a single pole. The shell is made of silver.

In analogy to Figure 3, Figure 4 shows the trajectories of the RS-eigenfrequencies, in this case depending on the core radius. In the subfigures (a1-a4), the real part of the eigenfrequencies are shown as a function of the core radius, while in subfigures (b1-b4) the trajectories in the complex plane are shown (the color gradient of the line encodes the core radius). Because of the spherical symmetry, distinct multipoles and polarizations are orthogonal to each other and can thus be treated individually. To enhance the readability and clarity of presentation, we restrict ourselves to the transverse magnetic dipolar resonances.

The same qualitative change in the trajectories from *weak* to *strong coupling* is clearly observed. Note how the metallic shell sustains two plasmon resonances; one predominantly at the outer surface and one mostly at the inner surface. The outer resonance (green line in (a); bright green line in (b)) is highly-damped due to strong radiative loss and couples only weakly to the core, due to a vanishing overlap. As such, it passes by without significantly interacting with the material resonance. The plasmon mode that lives mostly on the inner surface (blue line), on the other hand, exhibits trajectories very similar to the ones found in Figure 3, which equivalently permits the categorization into *weak*, *strong*, and *observable strong coupling* from the changing complex eigenfrequencies alone.

E. Multiple material resonances

Most systems of practical relevance contain materials that support more than one material resonance. When their resonance frequencies are well separated, their coupling to the cavity mode can be investigated individually. However, when the material resonances are located in close vicinity, separating their coupling behavior becomes non-trivial. Let us investigate the trajectories of the RSs of a planar cavity filled with a realistic 3-pole model of the SURMOF material [the dielectric function is shown in Figure 5(a)]. In this particular example, we do not adjust the coupling strength, but rather consider a single realistic material model: The electric dipolar polarizability of the unit cell of the SURMOF was quantum-

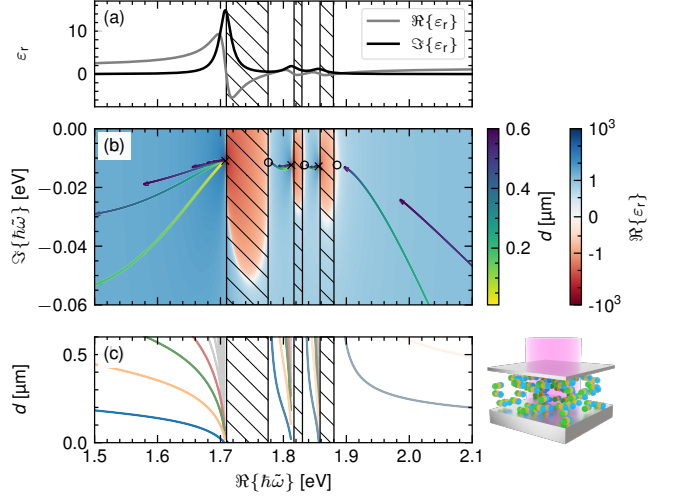


FIG. 5. Pole trajectories: A Fabry-Perot cavity made from two silver films separated by a varying thickness d is filled with a SURMOF material whose dielectric function can be described effectively by material resonances. (a) Real and imaginary part of the permittivity of the SURMOF material inside the cavity as extracted from quantum-chemical simulations. (b) Analogous to Figure 3(a), the trajectories of the complex eigenfrequencies of the RSs are parametrized by changing the cavity thickness d as indicated by their color. The background color corresponds to the real part of the SURMOF permittivity. The three material resonances (marked by x) generate one accumulation point each. With increasing cavity thickness, the modes dominated by the higher frequency material resonances leave the accumulation points, moving left towards the complex zeros of ϵ_r (open circles) created by the superposition of neighboring material resonances. (c) Projection of the pole trajectories onto the real frequency axis: The RSs are colored distinctly according to the Fabry-Perot mode they belong to. The dashed area marks the real frequency interval in which $\Re\{\epsilon\} < 0$.

chemically computed using the time-dependent density-functional theory. Subsequently, we applied a homogenization procedure to retrieve the dielectric function. Please see the SI for more details. The specific material parameters and geometry are again given in Table I of the SI. Figures 5(b) and (c) show, similarly to the previous figures, the eigenfrequencies of the RSs of the hybrid system in the complex plane (b) and the projection onto the real line (c) parametrized by the cavity thickness. Similar to the case of the single material resonance discussed above, the cavity mode is split into a lower and an upper polariton branch, which emerge from the lowest-energy material pole for a vanishing cavity thickness and approach the highest-energy material zero for the largest cavity thickness, respectively.

However, additional polariton branches emerge in-between the material resonances. These traverse the complex plane, emerging from different material resonances and approaching material zeros, but they never enter the region between the pole and the neighbor-

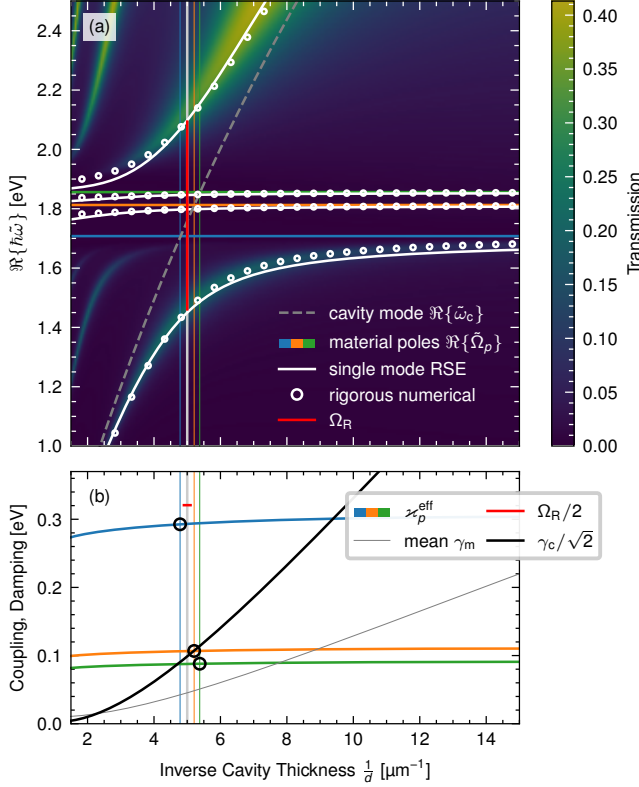


FIG. 6. Resonant state expansion introducing material resonances: (a) Starting from a single RS (the fundamental FP cavity mode - dashed line) and the three material resonances (solid colored lines), the resulting four resonance frequencies of the coupled system are recovered with remarkable accuracy (solid white lines). For reference, the corresponding RSs of the coupled system are calculated by solving the nonlinear eigenproblem numerically (circles), while the transmission of the cavity is obtained by conventional scattering simulations at real-valued frequencies (background). The red vertical line indicates the minimum splitting between the upper and lower polariton branches, which contains contributions from all three material resonances. (b) The effective coupling rates $\kappa_p^{\text{eff}} = \sqrt{|g_p \tilde{\Omega}_p|}$ (colored lines) computed from Equation (11) are compared to the average damping rate of all hybrid modes $\tilde{\omega}_m = \omega_m - i\gamma_m$ (solid black line). Further the scaled damping rate of the bare cavity mode $\gamma_c/\sqrt{2}$ (black line) is shown. The short red marker indicates the frequency splitting between the highest and lowest energy polariton branches extracted in (a).

ing higher-energy zero of any single material resonance. This behavior leads to energy gaps in the modal dispersion. Typically, these are associated with the medium inside the FP cavity turning effectively metallic ($\Re\{\varepsilon_r(\Re\{\omega\})\} < 0$; hatching in Figure 5) on parts of the real-frequency axis, thereby not supporting a FP mode. Note how the “keep out” regions are actually slightly larger, because the permittivity is now evaluated at complex frequencies.

Despite having access to the RSs of the coupled system, an interesting challenge remains: How can one investigate

the individual coupling of a cavity mode to one specific material resonance in the presence of additional material resonances?

The trajectories shown in Figure 5(b), generally exhibit a behavior that is similar to the previously considered example, when strong coupling is present. Due to the multiple material resonances, the behavior is richer in its details. The hybrid RSs once again swap their position. As before, we find the RS, that starts as the fundamental cavity mode and gets captured by the peripheral material zero, and the RS, that emerges from the lowest energy material resonance and asymptotically approaches the bare cavity mode. In addition we observe the pairwise exchange from higher-energy material poles (resonances) to material zeros. We conclude, that all of the material resonances are strongly coupled to the cavity mode, as none of the material poles is simply passed without being involved in these swaps.

However, evaluating the coupling strength beyond the weak coupling regime is more delicate. Simply considering the overall splitting (in red in Figure 6) provides no information on the individual coupling of the different material resonances.

To address this question, we derive an effective Hamiltonian $\hat{\mathcal{H}}$ that accounts for the hybridization between a single RS of the “empty” photonic resonator and an arbitrary number of material resonances (refer to Section IV C of the SI for details). Then, the hybrid light-matter RSs of the coupled system can be found as solutions to the eigenvalue problem:

$$\hat{\mathcal{H}}\psi = \tilde{\omega}\psi. \quad (10)$$

Here, $\tilde{\omega}$ are the hybrid eigenfrequencies, and the eigenvectors ψ contain the admixture coefficients of the original RS of the empty photonic resonator (first element) and the induced polarization of the individual material resonances (subsequent elements). For P material resonances supported by the material and embedded in the resonator, $\hat{\mathcal{H}}$ has the form

$$\hat{\mathcal{H}} = \begin{bmatrix} \tilde{\omega}_c - \sum_{p=1}^P g_p & \kappa_1 & \kappa_2 & \cdots \\ \hat{\kappa}_1 & \tilde{\Omega}_1 & 0 & \cdots \\ \hat{\kappa}_2 & 0 & \tilde{\Omega}_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

where $\hat{\kappa}_p = -\kappa_p = \sqrt{g_p \tilde{\Omega}_p}$, and g_p is the overlap between the optical mode and the p^{th} material residue:

$$g_p = \int_V \mathbf{E}_c(\mathbf{r}) \cdot i\sigma_p(\mathbf{r}) \mathbf{E}_c(\mathbf{r}) d\mathbf{r}. \quad (11)$$

The eigenproblem in Equation (10) exhibits great similarity to the coupled-oscillator model generalized to multiple material resonances. However, we obtain a correction to the eigenfrequency of the bare cavity mode $\tilde{\omega}_c$ given by $-\sum_p g_p$, which is absent in the phenomenological models.

Let us now demonstrate the applicability of our model to the example system previously introduced in Figure 5. A colormap of the observable transmission as a function of the inverse cavity thickness and the real frequency is shown in Figure 6. It is overlaid with resonance frequencies that were calculated in two ways: Given only the bare cavity RS [eigenfrequency (dashed line), normalized fields] and the material resonances [eigenfrequencies (colored lines), distribution in space $\sigma_p(\mathbf{r})$], the linear eigenproblem in Equation (10) predicts the hybrid eigenfrequencies (solid white lines). We compare these to numerical reference solutions (open white circles), obtained by directly solving the nonlinear eigenproblem posed by the cavity including all material resonances. Excellent agreement between the resonance frequencies in both of these approaches is found. As such we conclude, that the introduced resonant state expansion faithfully models the coupling between a photonic mode and multiple material resonances

Besides the accurate prediction of the hybrid eigenfrequencies, the method allows us to directly evaluate the overlaps g_p , giving us access to the coupling terms $\hat{\kappa}_p$ and κ_p without any fitting procedures. Let us now use these to investigate whether the strong coupling of individual material resonances is observable. In the presented example, $g_p\tilde{\Omega}_p$ is almost fully real. As such, we can restrict our analysis to individual effective retardation-free coupling rates

$$\kappa_p^{\text{eff}} = \sqrt{|g_p\tilde{\Omega}_p|}, \quad (12)$$

in analogy to the coupled-oscillator model. The three κ_p^{eff} from the current example are shown in Figure 6 (b).

The material resonance with the lowest frequency contributes significantly more to the overall splitting [red lines in both Figure 6(a) and (b)] than the remaining two resonances. Only the coupling of the dominant material resonance exceeds the damping of all bare resonances (light and matter). In particular, the coupling rates $\kappa_{2,3}^{\text{eff}}$ with the weaker material poles lie below γ_c in the relevant thickness range. The coupling rates of all three material resonances exceed the average damping of the four split polariton branches (gray line in Figure 6(b)). Considering the first form of the observable strong coupling criterion [Equation (8)], one might be misled to read this as an indication that all material resonances exhibit observable strong coupling. However, considering the average damping of all bare resonances is not a robust indicator insofar as it incorporates all far detuned and also very weakly coupled material resonances, playing a vanishing role in the system dynamics. Whether these irrelevant material resonances on average have high or low damping should play no role in the investigation of the coupling regime of a distinct material resonance.

On the other hand, *individually* comparing the effective coupling strength κ_p^{eff} to the damping rates of the uncoupled optical mode and the p^{th} material resonance provides a natural and robust way to generalize the second

criterion for observable strong coupling [Equation (9)] to multiple material resonances:

$$\kappa_p^{\text{eff}} > \sqrt{\frac{\gamma_c^2 + \gamma_p^2}{2}}. \quad (13)$$

Considering the low damping of the material resonances, *i.e.*, $\gamma_p \ll \gamma_c$, we introduce the following approximation

$$\sqrt{\frac{\gamma_c^2 + \gamma_p^2}{2}} \approx \frac{\gamma_c}{\sqrt{2}},$$

which is shown as the black line in Figure 6(b). If the material resonances had stronger damping we could simply proceed without the approximation, which would, however, clutter Figure 6(b). The simplified criterion now reads as

$$\kappa_p^{\text{eff}}(d) \gtrsim \frac{\gamma_c(d)}{\sqrt{2}}. \quad (14)$$

Here, we reintroduce the explicit dependence on the system parameter *rho*, which in this case corresponds to the cavity thickness d , previously omitted for readability. Due to the dispersive silver mirrors, $\gamma_c(d)$ changes significantly with d . Therefore, it is relevant to compare the coupling and damping for an appropriate d . Here, we chose the cavity thickness d_p such that the real parts of the bare cavity mode and p^{th} material resonance match (indicated by the colored vertical lines). The black circles in Figure 6(b) indicate the different $\kappa_p^{\text{eff}}(d_p)$, which have to be compared to $\gamma_c(d_p)/\sqrt{2}$ at the same thickness. From the generalized criterion above, we can conclude that the dominant material resonance (blue) exhibits observable strong coupling, while the second material resonance (orange) is on the verge of observable strong coupling, and for the third material resonance (green) the strong coupling is hidden by the damping of the optical cavity γ_c .

In summary, the presented framework enables the nuanced analysis of arbitrary combinations of material resonances with different couplings to a joint optical mode. While these calculations rely on overlap integrals of adequately normalized RSs, the individual coupling coefficients are phenomenologically accessible directly from the eigenfrequencies of the coupled system. A detailed explanation is provided in Section IV D of the SI.

III. CONCLUSION

In this work, we have shown that photonic RSs provide a rigorous and unambiguous framework for classifying regimes of light-matter interactions in open resonant systems. By following the trajectories of RSs in the complex frequency plane as a function of some system parameter, RSs reveal clear signatures of weak coupling, hidden strong coupling, and spectrally resolvable strong coupling. Importantly, this perspective captures subtle

effects often obscured in real-frequency observables, such as Purcell- and Lamb-like shifts, or the passage through exceptional points. They allow for direct extraction of coupling rates, even in the presence of multiple material resonances. While for spherical and planar systems, all necessary expressions can be derived from analytic equations [63–65], it is straight-forward to obtain them numerically in more complex systems [29, 43, 66].

Beyond the specific examples studied here, our study highlights the importance of considering the true eigenmodes of open photonic systems for designing and interpreting light–matter interactions in complex photonic environments. Hence, the RS framework provides a foundation to understand and design resonators for a broad range of applications in photonics and beyond, ranging from polariton lasing [4, 5] and computing [6–9] to polariton chemistry [14–16].

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Data Availability Statement — The datasets generated and analyzed during the current study are available in the github repository, www.github.com/tfp-photonics/agsurmof.

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- [1] C. Weisbuch, M. Nishioka, A. Ishikawa, and Y. Arakawa, *Physical Review Letters* **69**, 3314 (1992), publisher: American Physical Society.
 - [2] D. G. Lidzey, D. D. C. Bradley, M. S. Skolnick, T. Virgili, S. Walker, and D. M. Whittaker, *Nature* **395**, 53 (1998), publisher: Nature Publishing Group.
 - [3] S. ABERRA Guebrou, C. Symonds, E. Homeyer, J. C. Plenet, Y. N. Gartstein, V. M. Agranovich, and J. Bellessa, *Physical Review Letters* **108**, 066401 (2012), publisher: American Physical Society.
 - [4] L. V. Butov, *Nature* **447**, 540 (2007), publisher: Nature Publishing Group.
 - [5] C. Schneider, A. Rahimi-Iman, N. Y. Kim, J. Fischer, I. G. Savenko, M. Amthor, M. Lermer, A. Wolf, L. Worschech, V. D. Kulakovskii, I. A. Shelykh, M. Kamp, S. Reitzenstein, A. Forchel, Y. Yamamoto, and S. Höfling, *Nature* **497**, 348 (2013), publisher: Nature Publishing Group.
 - [6] D. Ballarini, M. De Giorgi, E. Cancellieri, R. Houdré, E. Giacobino, R. Cingolani, A. Bramati, G. Gigli, and D. Sanvitto, *Nature Communications* **4**, 1778 (2013), publisher: Nature Publishing Group.
 - [7] P. Tassan, D. Urbonas, B. Chmielak, J. Bolten, T. Wahlbrink, M. C. Lemme, M. Forster, U. Scherf, R. F. Mahrt, and T. Stöferle, *Integrated ultrafast all-optical polariton transistors* (2024), arXiv:2404.01868 [cond-mat, physics:physics].
 - [8] A. V. Zasedatelev, A. V. Baranikov, D. Urbonas, F. Scafirimuto, U. Scherf, T. Stöferle, R. F. Mahrt, and P. G. Lagoudakis, *Nature Photonics* **13**, 378 (2019), publisher: Nature Publishing Group.
 - [9] A. Kavokin, T. C. H. Liew, C. Schneider, P. G. Lagoudakis, S. Klemmt, and S. Hoeffling, *Nature Reviews Physics* **4**, 435 (2022), publisher: Nature Publishing Group.
 - [10] R. Mirek, A. Opala, P. Comaron, M. Furman, M. Król, K. Tyszk, B. Serebyński, D. Ballarini, D. Sanvitto, T. C. H. Liew, W. Pacuski, J. Suffczyński, J. Szczytko, M. Matuszewski, and B. Piętk, *Nano Letters* **21**, 3715 (2021), publisher: American Chemical Society.
 - [11] P. A. Thomas, K. S. Menghrajani, and W. L. Barnes, *Nature Communications* **13**, 1809 (2022), publisher: Nature Publishing Group.
 - [12] P. Zheng, S. Semancik, and I. Barman, *Nano Letters* **23**, 9529 (2023), publisher: American Chemical Society.
 - [13] R. Fei, M. P. Hautzinger, A. H. Rose, Y. Dong, I. I. Smalyukh, M. C. Beard, and J. van de Lagemaat, *The Journal of Physical Chemistry Letters* **15**, 1748 (2024), publisher: American Chemical Society.
 - [14] J. A. Hutchison, T. Schwartz, C. Genet, E. Devaux, and T. W. Ebbesen, *Angewandte Chemie International Edition* **51**, 1592 (2012).
 - [15] A. Thomas, L. Lethuillier-Karl, K. Nagarajan, R. M. A. Vergauwe, J. George, T. Chervy, A. Shalabney, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, *Science* **363**, 615 (2019), publisher: American Association for the Advancement of Science.
 - [16] M. Hertzog, M. Wang, J. Mony, and K. Börjesson, *Chemical Society Reviews* **48**, 937 (2019), publisher: Royal Society of Chemistry.
 - [17] V. Savona, L. C. Andreani, P. Schwendimann, and A. Quattropani, *Solid State Communications* **93**, 733 (1995).
 - [18] C. Carlson, R. Salzwedel, M. Selig, A. Knorr, and S. Hughes, *Physical Review B* **104**, 125424 (2021).
 - [19] B. Zerulla, D. Beutel, C. Holzer, I. Fernandez-Corbaton, C. Rockstuhl, and M. Krstić, *Advanced Materials* **36**, 2311405 (2024).
 - [20] M. F. Limonov, M. V. Rybin, A. N. Poddubny, and Y. S. Kivshar, *Nature Photonics* **11**, 543 (2017), publisher: Nature Publishing Group.
 - [21] S. Kéna-Cohen, M. Davanço, and S. R. Forrest, *Physical Review Letters* **101**, 116401 (2008), publisher: American Physical Society.
 - [22] B. Xiang, R. F. Ribeiro, M. Du, L. Chen, Z. Yang, J. Wang, J. Yuen-Zhou, and W. Xiong, *Science* **368**, 665 (2020), publisher: American Association for the Advancement of Science.
 - [23] A. Christ, S. G. Tikhodeev, N. A. Gippius, and

- H. Giessen, Phys. Rev. Lett. **91**, 183901 (2003).
- [24] A. Canales, T. Karmstrand, D. G. Baranov, T. J. Antosiewicz, and T. O. Shegai, Nanophotonics **12**, 4073 (2023).
- [25] E. A. Muljarov and W. Langbein, Physical Review B **93**, 075417 (2016), publisher: American Physical Society.
- [26] H. Sehmi, W. Langbein, and E. Muljarov, Physical Review B **101**, 045304 (2020), publisher: American Physical Society.
- [27] P. E. Stamatopoulou, *Strong Light-Matter Interactions in Extreme Plasmonic and Mie-Resonant Systems*, Springer Theses (Springer Nature Switzerland, Cham, 2024).
- [28] C. Hsu, B. Zhen, A. Stone, J. D. Joannopoulos, and M. Soljačić, Nat. Rev. Mater. **1**, 16048 (2016).
- [29] S. Both and T. Weiss, Semiconductor Science and Technology **37**, 013002 (2022).
- [30] C. Sauvan, T. Wu, R. Zarouf, E. A. Muljarov, and P. Lalanne, Optics Express **30**, 6846 (2022), publisher: Optica Publishing Group.
- [31] P. T. Kristensen, K. Herrmann, F. Intravaia, and K. Busch, Advances in Optics and Photonics **12**, 612 (2020), publisher: Optica Publishing Group.
- [32] B. Luk'yanchuk, N. I. Zheludev, S. A. Maier, N. J. Halas, P. Nordlander, H. Giessen, and C. T. Chong, Nature Materials **9**, 707 (2010), publisher: Nature Publishing Group.
- [33] A. E. Miroshnichenko, S. Flach, and Y. S. Kivshar, Reviews of Modern Physics **82**, 2257 (2010), publisher: American Physical Society.
- [34] C. Sauvan, J. P. Hugonin, I. S. Maksymov, and P. Lalanne, Physical Review Letters **110**, 237401 (2013), publisher: American Physical Society.
- [35] Q. Bai, M. Perrin, C. Sauvan, J.-P. Hugonin, and P. Lalanne, Optics Express **21**, 27371 (2013), publisher: Optica Publishing Group.
- [36] M. Fox, *Optical properties of solids*, 2nd ed., Oxford master series in condensed matter physics (Oxford Univ. Press, Oxford, 2012).
- [37] E. Hecht, *Optics* (Pearson Education India, 2012).
- [38] I. Ben Soltane, F. Dierick, B. Stout, and N. Bonod, Advanced Optical Materials **12**, 2400093 (2024).
- [39] D. Zheng, S. Zhang, Q. Deng, M. Kang, P. Nordlander, and H. Xu, Nano Letters **17**, 3809 (2017), publisher: American Chemical Society.
- [40] J. Wen, H. Wang, W. Wang, Z. Deng, C. Zhuang, Y. Zhang, F. Liu, J. She, J. Chen, H. Chen, S. Deng, and N. Xu, Nano Letters **17**, 4689 (2017), publisher: American Chemical Society.
- [41] M. S. Rider and W. L. Barnes, Contemporary Physics **62**, 217 (2021), publisher: Taylor & Francis.
- [42] P. Törmä and W. L. Barnes, Reports on Progress in Physics **78**, 013901 (2014), publisher: IOP Publishing.
- [43] P. Lalanne, W. Yan, K. Vynck, C. Sauvan, and J.-P. Hugonin, Laser & Photonics Reviews **12**, 1700113 (2018).
- [44] M. S. Rider, E. C. Johnson, D. Bates, W. P. Wardley, R. H. Gordon, R. D. J. Oliver, S. P. Armes, G. J. Leggett, and W. L. Barnes, Strong coupling in molecular systems: a simple predictor employing routine optical measurements (2024), arXiv:2402.09885 [physics].
- [45] L. Novotny and B. Hecht, *Principles of Nano-Optics*, 2nd ed. (Cambridge University Press, Cambridge, 2012).
- [46] W. L. Barnes, Strong coupling and the C=O vibrational bond (2023), arXiv:2307.02922 [physics].
- [47] K. S. Menghrajani, A. B. Vasista, W. J. Tan, P. A. Thomas, F. Herrera, and W. L. Barnes, The Journal of Physical Chemistry Letters **15**, 7449 (2024), publisher: American Chemical Society.
- [48] C. Schneider, M. M. Glazov, T. Korn, S. Höfling, and B. Urbaszek, Nature Communications **9**, 2695 (2018).
- [49] C. Symonds, C. Bonnand, J. C. Plenet, A. Bréhier, R. Parashkov, J. S. Lauret, E. Deleporte, and J. Bellessa, New Journal of Physics **10**, 065017 (2008).
- [50] S. Ghosh and T. C. H. Liew, npj Quantum Information **6**, 1 (2020), publisher: Nature Publishing Group.
- [51] J. G. Robertson, Publications of the Astronomical Society of Australia **30**, e048 (2013).
- [52] H. G. Schulze, S. Rangan, M. Z. Vardaki, M. W. Blades, R. F. B. Turner, and J. M. Piret, Applied Spectroscopy **76**, 61 (2022).
- [53] Y. Hagiwara and T. Kuwatani, Precision in peak parameter estimation for Gaussian and Lorentzian profiles: Guidelines for instrument optimization (2025).
- [54] O. P. Bruno, M. A. Santana, and L. N. Trefethen, Evaluation of resonances: adaptivity and AAA rational approximation of randomly scalarized boundary integral resolvents (2024), arXiv:2405.19582 [math].
- [55] Y. Nakatsukasa, O. Sète, and L. N. Trefethen, SIAM Journal on Scientific Computing **40**, A1494 (2018), publisher: Society for Industrial and Applied Mathematics.
- [56] F. Betz, M. Hammerschmidt, L. Zschiedrich, S. Burger, and F. Binkowski, Laser & Photonics Reviews **18**, 2400584 (2024).
- [57] J. D. Fischbach, F. Betz, N. Asadova, P. Tassan, D. Urbonas, T. Stöferle, R. F. Mahrt, S. Burger, C. Rockstuhl, F. Binkowski, and T. J. Sturges, Advanced Theory and Simulations **8**, 2400989 (2025).
- [58] A. Canós Valero, V. Bobrov, T. Weiss, L. Gao, A. S. Shalin, and Y. Kivshar, Physical Review Research **6**, 013053 (2024), publisher: American Physical Society.
- [59] T. J. Antosiewicz, S. P. Apell, and T. Shegai, ACS Photonics **1**, 454 (2014), publisher: American Chemical Society.
- [60] F. Stete, W. Koopman, C. Henkel, O. Benson, G. Kewes, and M. Bargheer, ACS Photonics **10**, 2511 (2023), publisher: American Chemical Society (ACS).
- [61] A. D. Kondorskiy, Optical Spectra of Plasmon-Exciton Core-Shell Nanoparticles: An Anisotropic Classical Model Eliminates Discrepancies with Experiments (2024), arXiv:2404.01718 [physics].
- [62] L. Greten, R. Salzwedel, T. Göde, D. Greten, S. Reich, S. Hughes, M. Selig, and A. Knorr, ACS Photonics **11**, 1396 (2024), publisher: American Chemical Society.
- [63] L. J. Armitage, M. B. Doost, W. Langbein, and E. A. Muljarov, Phys. Rev. A **89**, 053832 (2014).
- [64] M. B. Doost, W. Langbein, and E. A. Muljarov, Phys. Rev. A **90**, 013834 (2014).
- [65] J. Defrance and T. Weiss, Opt. Express **28**, 32363 (2020).
- [66] T. Weiss, M. Mesch, M. Schäferling, H. Giessen, W. Langbein, and E. A. Muljarov, Phys. Rev. Lett. **116**, 237401 (2016).

IV. SUPPLEMENTARY INFORMATION

A. Parameters of the investigated examples

The materials and resonator geometries in this article are inspired by the experimental and theoretical work presented in references [1] and [2], respectively. In those, a planar cavity is formed by depositing a film of surface-metal-organic framework (Zn-SiPc-SURMOF-2) between two thin silver layers, forming the top and bottom reflectors. While the optical properties of silver are readily available in the literature [3], the authors of [2] developed a multi-scale framework to obtain optical properties from quantum-chemical time-dependent density functional theory (TD-DFT) simulations, which are compatible with device-scale simulations. After obtaining molecular polarizabilities from the TD-DFT simulations, these are converted to dipolar transition matrices (T-matrices). The introduced framework then provides two distinct paths to compute the optical response of macroscopic objects. Either the T-matrices can be directly used in periodic and aperiodic multiple-scattering calculations via the transition matrix formalism. Alternatively, homogenized material parameters (such as the relative permittivity ε_r) can be obtained for periodic and disordered arrangements of molecules, which is the approach we follow here.

To enable evaluating the RSs, we further require material models that allow for analytic continuation to the complex frequency plane. Material models that are rooted in a description based on ordinary differential equations naturally fulfill this requirement. These include the above-mentioned Drude and Drude-Lorentz models. The homogenized permittivities of the SURMOF material were fit to good agreement with a three-pole Lorentz-oscillator model.

Geometrical and material parameters of the examples investigated throughout Figures 2 to 6 are provided in Table I at the end of the document.

B. Translating Lorentz to pole-based material model

Here, we provide the link between Eq. (3) and Eq. (4). For that purpose, the fractions on the right-hand side of Eq. (3) are expanded as a sum over simple poles as

$$\frac{f_j}{\omega_{0,j}^2 - \omega^2 - i\omega\Gamma_j} = \frac{A_j}{\omega - \Omega_{j,+}} + \frac{B_j}{\omega - \Omega_{j,-}}, \quad (15)$$

where $\Omega_{j,+} = \hat{\omega}_{0,j} - i\Gamma_j/2$ and $\Omega_{j,-} = -\Omega_{j,+}^*$. Defining $\hat{\omega}_{0,j}^2 \equiv \omega_{0,j}^2 - \frac{1}{4}\Gamma_j^2$, we find that B_j is given by

$$B_j = \frac{f_j}{2\hat{\omega}_{0,j}}, \quad (16)$$

and $A_j = -B_j$.

When further adding a background permittivity ε_∞ to encapsulate resonances outside the frequency domain of interest, Eq. (3) can be written as

$$\varepsilon_r(\omega) = \varepsilon_\infty + \quad (17)$$

$$\frac{Nq^2}{m_0\varepsilon_0} \sum_j \left(\frac{A_j}{\omega - \Omega_{j,+}} + \frac{B_j}{\omega - \Omega_{j,-}} \right). \quad (18)$$

Separating the two poles associated to each j , the latter expression takes the form of Eq. (4) in the main text:

$$\varepsilon_r(\mathbf{r}, \omega) = \varepsilon_\infty + \sum_p \frac{i\sigma_p}{\omega - \Omega_p}. \quad (19)$$

Following reference [4], we have introduced here the abbreviation σ_p as

$$\sigma_p = \mp \frac{Nq^2}{m_0\varepsilon_0} \frac{f_j}{2\hat{\omega}_{0,j}}, \quad (20)$$

with the $\mp$ signs corresponding to the $\Omega_{j,\pm}$ poles in Eq. (15), respectively.

C. Effective Hamiltonian in the single-mode approximation

We derive a first-principles theory on the hybridization of a single RS with an arbitrary number of material resonances. To do that, we start with the source-free Maxwell's curl equations in the frequency domain, which can be written in a compact notation as [5]:

$$\left[\frac{\omega}{c} \hat{\mathbb{P}}(\mathbf{r}; \omega) - \hat{\mathbb{D}} \right] \mathbb{F}(\mathbf{r}) = 0. \quad (21)$$

Here, we have introduced the material tensor $\hat{\mathbb{P}}(\mathbf{r}; \omega)$, assuming non-magnetic, achiral and isotropic materials, as:

$$\hat{\mathbb{P}}(\mathbf{r}; \omega) = \begin{bmatrix} \varepsilon_r(\mathbf{r}; \omega) \hat{I} & \mathbf{0} \\ \mathbf{0} & \hat{I} \end{bmatrix}. \quad (22)$$

In the above, $\hat{I}$ is the 3×3 identity matrix. The matrix $\hat{\mathbb{D}}$ is given by:

$$\hat{\mathbb{D}} = \begin{bmatrix} \nabla \times & \mathbf{0} \\ \mathbf{0} & \nabla \times \end{bmatrix}. \quad (23)$$

In addition, we have arranged the electric and magnetic fields in the vector $\mathbb{F}(\mathbf{r})$ as

$$\mathbb{F}(\mathbf{r}) = \begin{bmatrix} \mathbf{E}(\mathbf{r}) \\ iZ_0 \mathbf{H}(\mathbf{r}) \end{bmatrix}, \quad (24)$$

where Z_0 is the impedance of vacuum.

In this context, we are interested in understanding the interaction between an “empty” optical cavity filled with a material characterized by some host permittivity $\varepsilon_c(\omega)$, and the resonances of a target material that would be

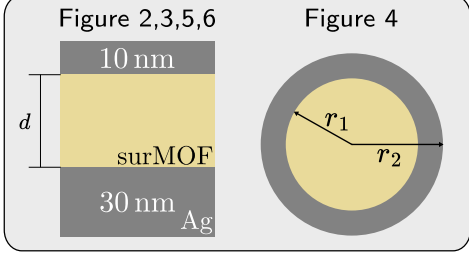
Figure	System	Geometry	Material Model surMOF	Material Model Silver
2,3	Planar Cavity	10 nm Silver variable cavity thickness d filled with surMOF 30 nm Silver	Drude Lorentz: 1. Pole: Oscillator Strength $\sqrt{f_1} = 2\pi \times 180.266\ 33\ \text{THz}$ Resonance $\omega_{0,1} = 2\pi \times 412.937\ 27\ \text{THz}$ Damping $\Gamma_1 = 2\pi \times 5.3\ \text{THz}$	Lorentz model: Oscillator Strength (or Plasma Frequency) $\sqrt{f_{\text{Ag}}} = 2\pi \times 1867.2715\ \text{THz}$ Damping $\Gamma = 2\pi \times 10.618\ 653\ \text{THz}$ Resonance $\omega_{0,\text{Ag}} = 2\pi \times 412.937\ 27\ \text{THz}$
4	Core Shell Sphere	variable $r_{\text{outer}} = r_2$ $r_1 = r_2 - 20\ \text{nm}$		
5,6	Planar Cavity	as Figure 1 	Additional Poles: 2. Pole: $\sqrt{f_2} = 2\pi \times 65.517\ 166\ \text{THz}$ $\omega_{0,2} = 2\pi \times 438.293\ 07\ \text{THz}$ $\Gamma_2 = 2\pi \times 6.0\ \text{THz}$ 3. Pole: $\sqrt{f_3} = 2\pi \times 53.923\ 443\ \text{THz}$ $\omega_{0,3} = 2\pi \times 448.791\ 10\ \text{THz}$ $\Gamma_3 = 2\pi \times 6.2\ \text{THz}$	

TABLE I. Geometry and material models of the treated examples.

introduced in the cavity. Upon filling the cavity with the material, the permittivity is modified by a series of P additional material poles:

$$\varepsilon_r(\mathbf{r}; \omega) = \varepsilon_c(\mathbf{r}; \omega) + \sum_{p=1}^P \frac{i\sigma_p(\mathbf{r})}{\omega - \tilde{\Omega}_p}, \quad (25)$$

Introducing Eq. (25) in Eq. (21), and moving the terms due to the new material to the right-hand side, we get:

$$\left[\frac{\omega}{c} \hat{\mathbb{P}}_c(\mathbf{r}; \omega) - \hat{\mathbb{D}} \right] \mathbb{F}(\mathbf{r}) = -\frac{\omega}{c} \sum_{p=1}^P \begin{bmatrix} \mathbf{P}_p(\mathbf{r}) \\ 0 \end{bmatrix}. \quad (26)$$

Here, $\mathbf{P}_p(\mathbf{r})$ is the polarization density induced by each new material pole on the cavity. The latter obeys the equation

$$(\omega - \tilde{\Omega}_p) \mathbf{P}_p(\mathbf{r}) = i\sigma_p \mathbf{E}(\mathbf{r}), \quad (27)$$

which is essentially the Fourier transform of the rate equation for the polarization density of the p^{th} material pole in the presence of an electric field.

Hence, the effect of filling the cavity with a dispersive material manifests as the emergence of polarization currents that perturb the eigenmodes of the empty cavity. The perturbed electric field can then be found with the electric part of the Green's function of the unperturbed system $\hat{\mathbf{G}}_c(\mathbf{r}, \mathbf{r}')$, by integrating over the volume V filled with the material:

$$\mathbf{E}(\mathbf{r}) = -\frac{\omega}{c} \sum_{p=1}^P \int_V \hat{\mathbf{G}}_c(\mathbf{r}, \mathbf{r}') \cdot \mathbf{P}_p(\mathbf{r}') d\mathbf{r}'. \quad (28)$$

We now assume the dynamics of the empty cavity are primarily driven by a single optical mode, with normalized electric field $\mathbf{E}_c(\mathbf{r})$ and eigenfrequency $\tilde{\omega}_c$. As a result, the Green's function can be approximately written as [5]:

$$\hat{\mathbf{G}}_c(\mathbf{r}, \mathbf{r}') \approx c \frac{\mathbf{E}_c(\mathbf{r}) \otimes \mathbf{E}_c(\mathbf{r}')}{\omega - \tilde{\omega}_c}. \quad (29)$$

In this case, $\otimes$ denotes the dyadic product. Inserting Eq. (29) in Eq. (28), we find that the perturbed eigenfield can be written as $\mathbf{E}(\mathbf{r}) = a \mathbf{E}_c(\mathbf{r})$, with the expansion coefficient a given by:

$$\begin{aligned} a &= - \sum_{p=1}^P \frac{\omega}{\omega - \tilde{\omega}_c} \int_V \mathbf{E}_c(\mathbf{r}) \cdot \mathbf{P}_p(\mathbf{r}) d\mathbf{r} \\ &= - \frac{\omega}{\omega - \tilde{\omega}_c} \sum_{p=1}^P n_p b_p. \end{aligned} \quad (30)$$

Here, we have introduced the abbreviation

$$b_p \equiv \frac{1}{n_p} \int_V \mathbf{E}_c(\mathbf{r}) \cdot \mathbf{P}_p(\mathbf{r}) d\mathbf{r}. \quad (31)$$

The coefficients b_p can be understood as the projection of the empty cavity field onto the polarization induced by the p^{th} additional material pole. Note how the unconjugated form of the projection is a direct consequence of the unconjugated field $\mathbf{E}_c(\mathbf{r})$ in the outgoing part of the Green's function in Eq. (29). The terms n_p are introduced to allow for an appropriate rescaling of the coefficients later on. Knowledge of the a and b_p coefficients is all that is needed to describe the degree of light-matter hybridization of the eigenmodes of the filled cavity.

However, we still require additional equations to determine the coefficients b_p . We first note that we can write Eq. (27) as:

$$(\omega - \tilde{\Omega}_p)\mathbf{P}_p = i\sigma_p a \mathbf{E}_c(\mathbf{r}). \quad (32)$$

Second, we take the scalar product of Eq. (32) with $\mathbf{E}_c(\mathbf{r})$ on the left and integrate over V to yield:

$$\omega n_p b_p = \tilde{\Omega}_p n_p b_p + g_p a. \quad (33)$$

The coupling term g_p takes the form of the overlap integral

$$g_p = \int_V \mathbf{E}_c(\mathbf{r}) \cdot i\sigma_p \mathbf{E}_c(\mathbf{r}) d\mathbf{r}. \quad (34)$$

We can now rewrite Eq. (30) and Eq. (33) as a linear eigenvalue problem for ω . For that purpose, we substitute the $\omega n_p b_p$ terms in Eq. (30) with the help of Eq. (33). After rearranging the resulting expression we get

$$\omega a = \left(\tilde{\omega}_c - \sum_{p=1}^P g_p \right) a - \sum_{p=1}^P \tilde{\Omega}_p n_p b_p. \quad (35)$$

Hereon, we make the replacement $\omega \rightarrow \tilde{\omega}$ to distinguish the resulting eigenfrequencies from an arbitrary frequency. The resulting system of equations can be written in matrix form:

$$\hat{\mathcal{H}}\psi = \tilde{\omega}\psi, \quad (36)$$

with the eigenvector ψ given by

$$\psi = \begin{bmatrix} a \\ b_1 \\ \vdots \\ b_P \end{bmatrix}, \quad (37)$$

and the “effective Hamiltonian” $\hat{\mathcal{H}}$:

$$\hat{\mathcal{H}} = \begin{bmatrix} \tilde{\omega}_c - \sum_{p=1}^P g_p & -n_1 \tilde{\Omega}_1 & -n_2 \tilde{\Omega}_2 & \cdots \\ g_1/n_1 & \tilde{\Omega}_1 & 0 & \cdots \\ g_2/n_2 & 0 & \tilde{\Omega}_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \quad (38)$$

The eigenvectors ψ quantify the degree of hybridization between the optical eigenmode and the material resonances. The strength of the interaction is determined by the off-diagonal elements of $\hat{\mathcal{H}}$.

Choosing $n_p = \sqrt{g_p/\Omega_p}$ we arrive at

$$\hat{\mathcal{H}} = \begin{bmatrix} \tilde{\omega}_c - \sum_{p=1}^P g_p & \kappa_1 & \kappa_2 & \cdots \\ \hat{\kappa}_1 & \tilde{\Omega}_1 & 0 & \cdots \\ \hat{\kappa}_2 & 0 & \tilde{\Omega}_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad (39)$$

with $\hat{\kappa}_p = -\kappa_p = \sqrt{g_p \Omega_p}$.

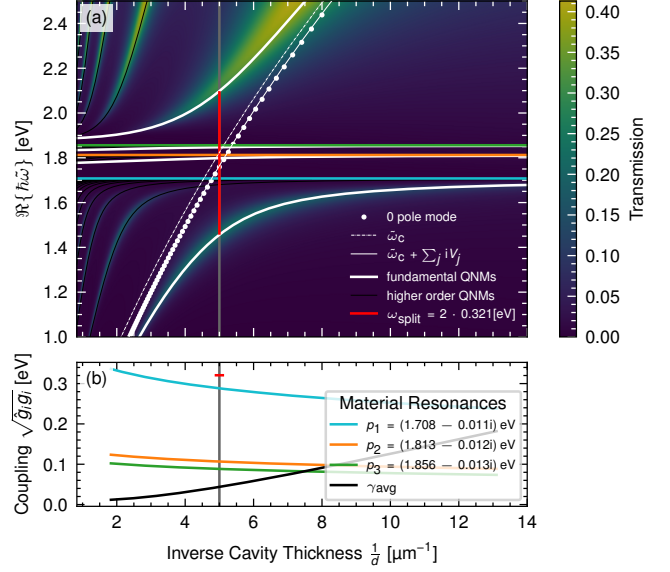


FIG. 7. Extracting the coupling coefficients from the complex RS frequencies: (a) We only consider RSs associated to the fundamental Fabry-Perot resonance. The cavity thickness at which the separation between the upper- and lowermost RS is minimized is indicated by the red line. This minimum separation corresponds to the Rabi frequency typically extracted from experiments. (b) Coupling rates between the optical and material poles, which was extracted from the RS trajectories by solving the inverse eigenproblem posed by Eq. (40). The short red marker indicates the Rabi frequency extracted in (a).

D. Inverse Eigenproblem

Here, we show how the coupling rates can be found phenomenologically in the presence of more than one material resonance if the eigenfrequencies of the coupled system are known. Let us start from the effective Hamiltonian obtained above (let us exemplarily use 3 material resonances):

$$\begin{bmatrix} \dot{\Phi}_c \\ \dot{\Phi}_1 \\ \dot{\Phi}_2 \\ \dot{\Phi}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} \tilde{\omega}_c - K & \kappa_1 & \kappa_2 & \kappa_3 \\ \hat{\kappa}_1 & \tilde{\Omega}_1 & 0 & 0 \\ \hat{\kappa}_2 & 0 & \tilde{\Omega}_2 & 0 \\ \hat{\kappa}_3 & 0 & 0 & \tilde{\Omega}_3 \end{bmatrix}}_{\mathcal{H}_3} \begin{bmatrix} \Phi_c \\ \Phi_1 \\ \Phi_2 \\ \Phi_3 \end{bmatrix}. \quad (40)$$

Here, we have summarized the correction to the eigenfrequency of the cavity mode as K . $\mathcal{H}$ must have eigenvalues corresponding to the resonance frequencies of the coupled system, which can be found directly by solving the nonlinear eigenproblem posed by the sourceless Maxwell equations with outgoing boundary conditions. As the characteristic polynomial contains contributions from $\hat{\kappa}_p$ and κ_p only in the products $\hat{\kappa}_p \kappa_p$, we can consider those products as our free variable instead. This leaves us with four unknowns. From the four known eigenvalues, we can construct a system of four equations (*e.g.*, for the coef-

ficients in the polynomial). As a consequence, we can unambiguously solve for the unknowns $\hat{\kappa}_p \kappa_p$ and $\tilde{\omega}_c$.

Let us write this out step by step. The characteristic polynomial is defined as:

$$C(\lambda) = \begin{vmatrix} \tilde{\omega}_c - K - \lambda & \kappa_1 & \kappa_2 & \kappa_3 \\ \hat{\kappa}_1 & \tilde{\Omega}_1 - \lambda & 0 & 0 \\ \hat{\kappa}_2 & 0 & \tilde{\Omega}_2 - \lambda & 0 \\ \hat{\kappa}_3 & 0 & 0 & \tilde{\Omega}_3 - \lambda \end{vmatrix}. \quad (41)$$

Its zeros correspond to the eigenvalues of the Hamiltonian, which in turn correspond to the RS frequencies, which can be derived analytically or numerically. We use the second row to develop the determinant as:

$$\begin{aligned} C(\lambda) &= (\tilde{\Omega}_1 - \lambda) \underbrace{\begin{vmatrix} \tilde{\omega}_c - K - \lambda & \kappa_2 & \kappa_3 \\ \hat{\kappa}_2 & \tilde{\Omega}_2 - \lambda & 0 \\ \hat{\kappa}_3 & 0 & \tilde{\Omega}_3 - \lambda \end{vmatrix}}_{C_{\setminus \tilde{\Omega}_1}(\lambda)} \\ &\quad - \hat{\kappa}_1 \begin{vmatrix} \kappa_1 & \kappa_2 & \kappa_3 \\ 0 & \tilde{\Omega}_2 - \lambda & 0 \\ 0 & 0 & \tilde{\Omega}_3 - \lambda \end{vmatrix} \\ &= (\tilde{\Omega}_1 - \lambda) C_{\setminus \tilde{\Omega}_1}(\lambda) - \hat{\kappa}_1 \kappa_1 \begin{vmatrix} \tilde{\Omega}_2 - \lambda & 0 \\ 0 & \tilde{\Omega}_3 - \lambda \end{vmatrix} \\ &= (\tilde{\Omega}_1 - \lambda) C_{\setminus \tilde{\Omega}_1}(\lambda) - \hat{\kappa}_1 \kappa_1 (\tilde{\omega}_{x,2} - \lambda)(\tilde{\omega}_{x,3} - \lambda). \end{aligned} \quad (42)$$

Here, $C_{\setminus \tilde{\Omega}_1}(\lambda)$ is the characteristic polynomial for the system after eliminating $\tilde{\Omega}_1$. Resolving this recursion, we can write down the characteristic polynomial for an arbitrary number of material resonances explicitly as:

$$\begin{aligned} C(\lambda) &= \underbrace{\left[\prod_p (\tilde{\Omega}_p - \lambda) \right]}_{\equiv R(\lambda)} (\tilde{\omega}_c - K - \lambda) \\ &\quad - \sum_p \hat{\kappa}_p \kappa_p \underbrace{\prod_{j \setminus p} (\tilde{\Omega}_j - \lambda)}_{\equiv Q_p(\lambda)}. \end{aligned} \quad (43)$$

From the definitions of $R(\lambda)$ and $Q_p(\lambda)$, we can derive:

$$Q_i(\lambda) = R(\lambda) / (\tilde{\omega}_{x,i} - \lambda). \quad (44)$$

We can now use these expressions to write down the system of equations resulting from the different eigenvalues λ_l compactly as

$$C(\lambda_l) = R(\lambda_l) \left[\tilde{\omega}_o - \lambda_l - \sum_i \frac{\hat{\kappa}_p \kappa_p}{\tilde{\omega}_{x,i} - \lambda_l} \right] = 0. \quad (45)$$

When knowing λ_l and by using $R(\lambda_l) \neq 0$, the remaining system of equations is easily solved.

The results are shown in Figure 7. A python implementation to solve the inverse eigenproblem is provided in the accompanying repository [6].

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- [1] R. Haldar, Z. Fu, R. Joseph, D. Herrero, L. Martín-Gomis, B. S. Richards, I. A. Howard, A. Sastre-Santos, and C. Wöll, *Chemical Science* **11**, 7972 (2020), publisher: The Royal Society of Chemistry.
 - [2] B. Zerulla, M. Krstić, D. Beutel, C. Holzer, C. Wöll, C. Rockstuhl, and I. Fernandez-Corbaton, *Advanced Materials* **34**, 2200350 (2022).
 - [3] P. B. Johnson and R. W. Christy, *Physical Review B* **6**, 4370 (1972), publisher: American Physical Society.
 - [4] H. Sehmi, W. Langbein, and E. Muljarov, *Physical Review B* **101**, 045304 (2020), publisher: American Physical Society.

- [5] T. Weiss and E. A. Muljarov, *Physical Review B* **98**, 085433 (2018), publisher: American Physical Society.
- [6] J. D. Fischbach, S. Gladyshev, A. Canós Valero, M. Nyman, T. Weiss, and C. Rockstuhl, *Source Code and Simulation Results: Resonant states reveal strong light-matter coupling in nanophotonic cavities* - [www.github.com/tfp-photonics/agsumof](https://github.com/tfp-photonics/agsumof) (2025).